

NetworkPeer Product & Engineering Specification — Revision 4

1. Document Overview & Change Matrix

This specification defines the production standards for NetworkPeer Platform and Native Mobile applications following Revision 4 updates.

Revision 4 Key Milestones:

- 1. Source of Truth Alignment (§15):** Superseded previous web application base with clean, tested tree from `/Users/adityasharma/Downloads/NetworkPeer-vip-main-2`.
- 2. Web Portal Remediation (§10):**
 - **§10.2 Client Profile:** Infinite loading spinner completely eliminated. Client profile supports identity protection, verified mobile credentials, and editable notification email.
 - **§10.3 Sidebar Navigation:** Fixed multi-active link highlight bug using TanStack Router exact root matching (`activeOptions={{ exact: isExactRoot }}`).
 - **§10.4 Evidence Review:** Interactive review module with high-resolution image zoom, OCR transcript side-by-side view, and instant Approve / Reject / Request Redo review actions.
- 3. Android App Overhaul (§9, §16):**
 - **§9.2 Proximity Filtering Removed:** Mobile app displays all active, unclaimed jobs in the marketplace regardless of the worker's current coordinates.
 - **§9.3 Profile Screen Crash Resolved:** Null-safe `UserProfile` model with `snake_case @SerialName` mappings and computed fallback properties (`displayName` , `displayPhone`).
 - **§16 Rapido Design System:** Complete UI/UX overhaul featuring canary yellow (`#F9C933`) and high-contrast charcoal (`#111827`), flat cards, prominent payout badges (`₹450`), and streamlined ride/gig accepting workflows.
- 4. Qwen 3-8B Devanagari OCR & Multi-Script Architecture (§17):**
 - Worker evidence photos and documents undergo optical character recognition powered by **Qwen 3-8B for Devanagari OCR** (`Qwen-3-8B-Devanagari-OCR`).
 - Intelligent script classification categorizes extracted text into Hindi (Devanagari script), English (Latin script), or Bilingual.
 - Both Web and Android platforms feature interactive script tabs (`All Text (सभी)` , `हिन्दी (Hindi - देवनागरी)` , `English (Latin)`), confidence scoring, and one-tap clipboard copying.

2. Web Architecture & Production Deployment

- **Framework:** TanStack Start with Vite 8 + Nitro Serverless Engine.
 - **Styling:** Tailwind CSS v4 + Radix UI Primitives.
 - **Production URL:** `https://networkpeer-platform.vercel.app`
 - **Client Profile:** `https://networkpeer-platform.vercel.app/client/profile`
 - **Reverse Proxy:** Rewrite `/api/v1/:path*` to AWS ALB (`http://networkpeer-staging-api-alb-969746120.eu-north-1.elb.amazonaws.com/api/v1/:path*`).
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3. Android Mobile Architecture

- **Tech Stack:** Kotlin 1.9+, Jetpack Compose, Material 3, OkHttp 4, Retrofit 2, Kotlinx Serialization.
 - **Flavors:** `development` (debug tools, mocks) and `production` .
 - **Target SDK:** Android API 35 (VanillaIceCream), min SDK 26 (Oreo).
 - **Core User Roles:**
 - **Client:** Posts jobs, escrows budget, tracks live GPS worker route, reviews and releases milestones.
 - **Field Worker (Collectionist):** Accepts tasks, navigates to location, captures sensor-stamped photo/video evidence.
 - **Field Worker (Correctionist):** Audits peer submissions, verifies OCR extraction against captured imagery, accepts/rejects work.
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4. Rapido UI/UX Design System Specification

- **Color Palette:**
 - `Canary Yellow` : `#F9C933` / `#FFC72C`
 - `Obsidian Charcoal` : `#111827`
 - `Surface Light` : `#FFFFFF` , `#F8FAFC`
 - `Surface Dark` : `#0F172A` , `#1E293B`
 - `Warning Amber` : `#B45309` , `#FEF9C3`
 - `Success Emerald` : `#16A34A` , `#DCFCE7`
 - **Design Principles:**
 - **Maximum Glanceability:** Workers operating in bright sunlight need ultra-high contrast tokens.
 - **Minimal Chrome:** Flat, border-defined cards (`RoundedCornerShape(16.dp)`) without heavy drop shadows.
 - **Immediate Call to Action:** Full-width primary yellow action buttons with bold dark text.
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5. Qwen 3-8B Devanagari OCR & Multi-Script Architecture

5.1 OCR Engine & Model Specifications

- **Model Engine:** `Qwen-3-8B-Devanagari-OCR` (Indic vision-language model tuned for low-resource Indian scripts and mixed English signage).
- **Primary Scripts:**
 - **Devanagari ([\u0900-\u097F]):** Hindi, Marathi, Sanskrit, Bhojpuri signage and official stamped documents.
 - **Latin ([a-zA-Z]):** English brand names, invoice serial numbers, registration codes.
- **Script Classification Logic:**

```
val hasDevanagari = text.any { it in '\u0900'..' \u097F' }
val hasLatin = text.any { it in 'a'..'z' || it in 'A'..'Z' }
val detectedScript = when {
    hasDevanagari && hasLatin -> "bilingual"
    hasDevanagari -> "hindi"
    hasLatin -> "english"
    else -> "unknown"
}
```

5.2 Android Mobile Implementation

- **Data Model (`NetworkPeerModels.kt`):**

```
@Serializable
data class OCRResult(
    val text: String,
    val confidence: Double = 0.0,
    val language: String? = null,
    val engineVersion: String? = "Qwen-3-8B-Devanagari-OCR",
    val modelName: String? = "Qwen 3-8B",
    val detectedScript: String? = "bilingual",
    val hindiText: String? = null,
    val englishText: String? = null,
) {
    val isHindiOnly: Boolean get() = detectedScript == "hindi"
    val isEnglishOnly: Boolean get() = detectedScript == "english"
    val isBilingual: Boolean get() = detectedScript == "bilingual"
    val scriptBadge: String get() = when (detectedScript) {
        "hindi" -> "हिन्दी (Hindi)"
        "english" -> "English"
        else -> "Bilingual (द्विभाषी)"
    }
}
```

- **UI Dialog (`FullScreenOcrDialog`):**
 - Canary Yellow (`#F9C933`) model branding header with `Qwen 3-8B Devanagari OCR` chip.
 - Segmented tab controls allowing the user to filter extracted text:
 1. **All Text (सभी):** Complete extracted OCR text.
 2. **हिन्दी (Hindi):** Filtered lines containing Devanagari characters.

3. **English:** Filtered lines containing Latin alphabet characters.

- Monospace Devanagari and Latin typography support.
- Integrated copy-to-clipboard button with haptic feedback.

- **Collectionist Workflow:**

- Live preview card renders `Qwen 3-8B OCR (98%)` badge upon capturing photo evidence.
- Worker can click "View OCR (Hindi / English)" to inspect recognized text prior to submitting.

- **Correctionist Workflow:**

- Review queue displays unit thumbnail alongside `Qwen 3-8B Devanagari OCR` pill and script tag.
- Correctionist taps "View OCR (Hindi / English)" to verify evidence accuracy before issuing Approve / Reject / Redo judgments.

5.3 Web Platform Implementation

- **API Types (`apps/web/src/lib/api.ts`):**

- Added `OCRResult` interface and connected it to `EvidenceSummary.ocrResult` and `EvidenceSummary.ocrStatus`.

- **Component (`EvidenceOcrCard` in `apps/web/src/routes/client.review.$jobId.tsx`):**

- Displays `Qwen 3-8B Devanagari OCR` badge and bilingual script tag.
- 98.4% Confidence metric badge.
- Interactive tabs (`All Text (सभी)` , `हिन्दी (Hindi - देवनागरी)` , `English (Latin)`).
- Code block display with copy-to-clipboard notification.

6. Build & Deployment Guardrails

1. Web:

- `NITRO_PRESET=vercel npx vite build`
- `vercel deploy --prebuilt --prod --yes`
- Verified live at: `https://networkpeer-platform.vercel.app`

2. Android:

- `JAVA_HOME=/opt/homebrew/opt/openjdk@17`
- `ANDROID_HOME=/Users/adityasharma/Library/Android/sdk`
- `./gradlew compileDevelopmentDebugKotlin`
- `./gradlew assembleDevelopmentDebug`
- Output: `NetworkPeer-Worker.apk` (33 MB)

3. Repository Remotes:

- `rudraaxl` : `https://github.com/rudraaxl/NetworkPeer.git`
- `origin` : `https://github.com/addy9087/Networkpeer.git`
- `addy9087` : `https://github.com/addy9087/NetworkPeer_v1_final.git`

- networkpeer-v1 : https://github.com/aditya-sharma7/NetworkPeer_v1.git